

THE FLOOD AND CHALLENGES OF RASUWAGADHI, KERUNG BORDER

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ABSTRACT

This article concerns a glacial lake outburst flood (GLOF) that took place on 8 July 2025 from Puruogangri Glacier in Tibet, which particularly impacted Rasuwagadhi, one of the most significant customs checkpoints on the border of Nepal and China. The outburst flooding led to the destruction of crucial infrastructure such as the Nepal-China Friendship Bridge, highways, and four hydropower plants, resulting in a loss of almost 9% of the total electricity generation capacity of Nepal. Moreover, the disaster claimed the lives of at least nine individuals and left nineteen others missing. The paper investigates the available meteorological information, satellite imagery, and analysis of secondary disaster reports to understand the impacts and the primary reasons behind the phenomenon. The analysis indicated that the catastrophic event was not a result of high abnormal precipitation, but rather, was triggered by rapid glacial melting and the sudden growth of a supraglacial lake during the 2024-2025 period. Cross-sectional data from GLOF events in the Himalayas in Nepal, in India, and Peru demonstrates the recurring and cross-border characteristics of GLOF events in high mountain regions. The paper outlines the need for proper management of the effects of climate change as it concerns bolstering glacier hazards in addition to which, 'systematic monitoring', 'international collaboration', and 'framework risk mitigation' must not be overlooked.

Keywords: *Glacial Lake Outburst Flood (GLOF), Himalayan cryosphere, Climate change impacts, Hydropower vulnerability, Transboundary water governance, Disaster risk reduction, Remote sensing, High Mountain hazards*

1. BACKGROUND

Rasuwagadhi is known as a major Himalayan customs checkpoint located in Rasuwa district of Nepal. It lies along the Rasuwagadhi - Kerung road, which connects Kathmandu with Tibet directly. This road serves as an important trade route between Nepal and China. Local communities live in the hilly terrain at an elevation of about 2,800 to 3,000 meters, sustained by the Bhotekoshi River that gives life to the region.

As Nepal's second official border checkpoint with China, this location carries both economic and geopolitical importance (Figure I shows this border before the devastating flood). The Lhunde and Kerung rivers of Tibet are the main sources shaping the Bhotekoshi Valley. Since these Himalayan rivers flow with great force, the risk is ever-present.

2. LITERATURE REVIEW

Glacial lake outburst floods (GLOFs) are a well-documented consequence of glacier retreat and shifting cryospheric dynamics, driven by increased meltwater and changes in permafrost and debris cover (Ives, Shrestha, & Mool, 2010). Foundational research established hazard contexts and risk assessment methods for the Himalayas (Khanal, Mool, Shrestha, & Bajracharya, 2015), while broader socioenvironmental studies show how glacier change

intersects with local livelihoods and governance (Carey, 2010). Studies of permafrost thaw and landslides highlight cascading dynamics in which slope failures and rock–ice avalanches can either form new lakes or directly trigger catastrophic floods, as seen in the Chamoli disaster (Shugar, et al., 2021) (Huggel, Clague, & Korup, 2012). Remote sensing and satellite imagery are now central for monitoring lake development and mapping post-event impacts (GoogleEarth, 2025) (ICIMOD, International Centre for Integrated Mountain Development, 2020), yet inventories emphasize that many potentially dangerous features remain under-observed without systematic and frequent monitoring.



Figure 1: Kerung Border before flood, Source: *(RSS, Rastriya Samachar Samiti, 2023)*

The 2025 Rasuwagadhi flood underscores these gaps. Preliminary analyses attribute the event to supraglacial lake drainage and note rapid, transboundary impacts on infrastructure and communities (TheKathmanduPost, 2025). Despite the involvement of national agencies and forecasting institutions, the disaster highlighted challenges of short warning times, limited cross-border data sharing, and roadside cargo staging that exacerbated exposure. Taken together, the literature points to four urgent priorities: expand monitoring of debris-covered glaciers and supraglacial ponds, integrate remote sensing with real-time forecasting and community alert systems, apply established GLOF risk assessment frameworks to supraglacial and cascading triggers, and strengthen investment in prevention and resilient infrastructure design (Ives, Shrestha, & Mool, 2010) (Khanal, Mool, Shrestha, & Bajracharya, 2015) (ICIMOD, Preliminary analysis of the Rasuwagadhi flood: Evidence of supraglacial lake drainage, 2025).

3. OBJECTIVES

This report is prepared with the intention of meeting a number of major goals, outlined below:

1. To discuss the causes and effects of the July 8, 2025 Rasuwagadhi flood.
2. To discuss the impact of climate change and review the event in relation to other GLOFs in Nepal and elsewhere.

3. To advise essential steps to reduce disaster risk through tracking, preparation, and collaboration.

4. RESEARCH GAP AND PROBLEM STATEMENT

The lack of preparedness towards GLOFS while considering it one of the most dangerous climatic threats in the Himalayas is shocking. The Rasuwagadhi flood is an excellent example that shows the extent to which rapidly draining supraglacial ponds can destroy bridges, hydropower plants and trade routes in a matter of hours. Very few infrastructures in Nepal have GLOF risks incorporated in their planning.

Many lakes that are located in Tibet and floods that occur in Nepal downwards are a result of the lack of coordinated relations between Nepal and China. This impacts the flood preparedness as the warning and response mechanisms are not coordinated. The other issue is that while supraglacial ponds are dangerous, monitoring still only focuses on larger moraine-dammed lakes.

The lack of understanding of the economic and environmental costs associated with GLOF is alarming. The potential losses are higher than the cost that needs to be incurred for the preventive measures. However, the comparison is absent which leads the government to not spend money for the long-term plan. There is a strong need of research in cross-border electronic real time monitoring, resilient infrastructure, shared border data, and low-cost strategies on risk reduction.

5. MATERIALS AND METHODS

All the information and data applied in the current paper were collected through secondary sources, and these are peer-reviewed journal publications, reports, and news articles regarding Glacial Lake Outburst Floods (GLOFs) in the Himalayas and concentrating chiefly on the Rasuwagadhi - Bhotekoshi section. Information on flood events, lake and river behavior and observation networks and mitigation actions were abstracted, collated and enriched by information gathered through consultations with colleagues and experts.

6. CASE STUDY

6.1 THE FLOOD OF JULY 8, 2025

Around 3:15 a.m. on July 8, 2025, a sudden and massive flood struck on the Bhotekoshi River. The flood destroyed the Nepal - China Friendship Bridge, swept away customs infrastructure, damaged roads, and caused severe losses to hydropower projects. Hundreds of trucks, electric vehicles, and heavy machinery were reported to have been carried away by the river. Storage warehouses were destroyed, and commercial activities came to a halt (Figure 2 illustrates the Kerung border after the flood).

This year's monsoon caused major damage along the Galchi - Trishuli - Rasuwagadhi highway. Continuous rainfall triggered landslides, damaging over one kilometer of the highway. Local bridges and slopes along this trade route, which connects Nepal with Tibet, also collapsed. As a result, trade activities between the two countries, as well as local movement, were disrupted. The damage not only halted commerce but also severed contact with the border area.

In terms of human loss, at least nine Nepalis were killed and nineteen went missing, including some Chinese nationals (Nepal Disaster Risk Reduction Authority, 2025).



Figure 2: Kerung Border after flood, Source: (RSS, Rastriya Samachar Samiti, 2025)

The initial rescue managed to bring more than fifty people to safety. Four major hydropower projects were affected: Rasuwagadhi (111 MW), Trishuli - 3A (60 MW), Trishuli (21 MW), and Mailung (15 MW). This resulted in a loss of about 8 - 9% of the national power generation capacity (Nepal Electricity Authority, 2025). The Prime Minister was compelled to declare a state of emergency.

6.2 CAUSE OF THE FLOOD

The flood was not related to rainfall. Satellite images and meteorological data confirmed that there had been no unusual precipitation (NOAA, 2025); (ECMWF, 2025). However, the water level in the upper Bhotekoshi region suddenly rose by 3 - 4 meters, which is typically a sign of a glacial lake outburst. Research indicates that the flood (GLOF - Glacial Lake Outburst Flood) was triggered by the bursting of a supraglacial lake formed in the Puruogangri Glacier in Tibet. Due to abnormal temperatures, the lake had expanded rapidly in 2024 - 25 and eventually burst on July 8 (ICIMOD, Preliminary analysis of the Rasuwagadhi flood: Evidence of supraglacial lake drainage, 2025).

7. DISCUSSIONS

The following discussion explores historical incidents, the danger today, and key mitigation strategies for Glacial Lake Outburst Floods (GLOFs) in the Himalayas and sheds light on key lessons in monitoring, infrastructure, community preparation, and trans-frontier cooperation.

7.1 HISTORY AND COMPARISON

In Nepal, the Dig Tsho glacial lake outburst in 1985 swept away hydropower projects and hundreds of homes (Ives, Shrestha, & Mool, 2010). In 2021, a flash flood in Chamoli, India, claimed more than 200 lives (Shugar, et al., 2021). Similarly, in Peru, a glacial lake outburst in 1941 caused thousands of deaths (Carey, 2010). These examples show that such risks are not limited to the Himalayas but are shared across other mountain regions as well.

7.2 CLIMATE CHANGE AND RISK

Research has shown that the average temperature in the Himalayas is rising at twice the global rate. This has caused glaciers to melt rapidly, forming new lakes. According to ICIMOD, there are currently about 5,000 glacial lakes in Nepal, which is 14% more than in 1990 (ICIMOD, International Centre for Integrated Mountain Development, 2020). This trend is increasing the risk of floods in the coming years.

7.3 LESSONS AND PREPAREDNESS: MANAGING GLACIAL LAKE RISKS

The flood in the Rasuwagadhi region has highlighted the emerging climate risks in the Himalayas and the challenges of managing them. The key lessons observed from this event are as follows:

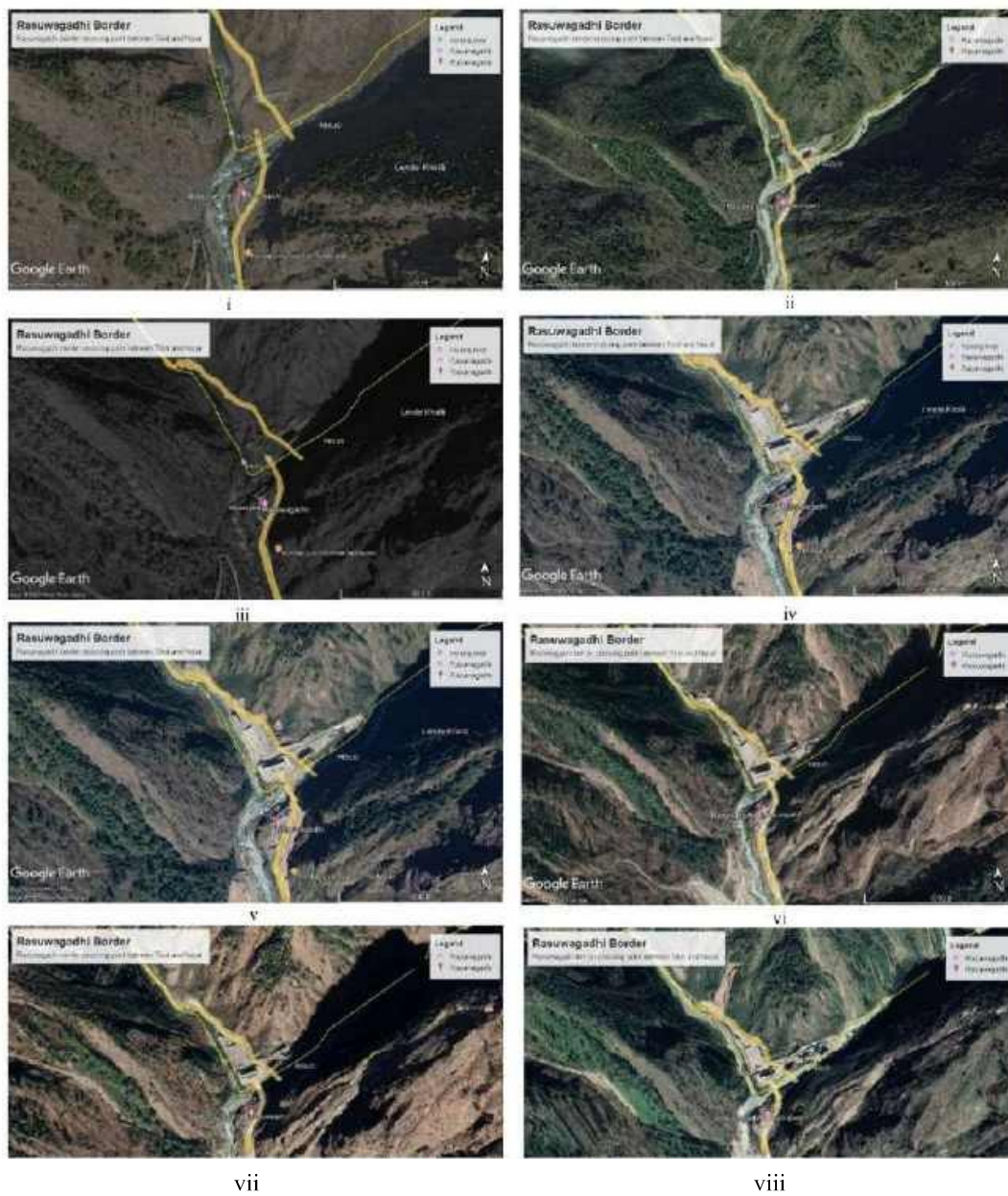


Figure 3: Satellite images of Kerung border area from 2005 to 2021. i.e. (i) 2005, (ii) 2011, (iii) 2012, (iv) 2014, (v) 2015, (vi) 2017, (vii) 2020, (viii) 2021 Source: (GoogleEarth, 2025)

7.4 MONITORING AND EARLY WARNING SYSTEMS

Glacier lakes are constantly dynamic. Melting snows, warmer temperatures, and unstable geology equate to fast shifts in pressure and in levels. Continuous monitoring through satellite imagery, aerial photography, and ground-based sensors is therefore essential. If this data is collected in real time and linked to local warning systems, communities at risk can be alerted on time, helping reduce both human and material losses.

7.5 LAKE MANAGEMENT

The prevention of sudden glacier lake outburst floods (GLOFs) is by carrying out controlled release of the water. This is achieved by the use of artificial channels, intake piping systems, or siphoning systems. Such management helps balance water pressure within the lake and reduces the risk of sudden outflows.

7.6 INFRASTRUCTURE STRENGTHENING

Mountain sites are vulnerable to risk in road and bridge ventures and hydropower. To make these structures sustainable in the long term, additional safety standards must be incorporated into their design. For example, bridges should be built to withstand maximum possible flood levels, road drainage systems should be improved, and hydropower plants should include emergency spillways.

7.7 COMMUNITY PREPAREDNESS

Technical systems alone are not enough; active community participation is crucial in risk management. Local residents must be equipped with emergency response plans, safe site identification, and regular rescue drills. Training schools, women's groups, and local organizations can make disaster response systems faster and more effective.

7.8 CROSS-BORDER COOPERATION

Glacial lakes and their risks are not confined to a single country's boundaries. For Himalayan nations like Nepal and China, data sharing, joint monitoring systems, and rapid exchange of emergency information can take risk management to a new level. Such cooperation strengthens scientific research while also creating an environment for immediate assistance during emergencies.

8. CONCLUSION

The Rasuwagadhi flood from July, 8, 2025 was a result of the sudden bursting of a supraglacial lake which was formed over the Puruogangri Ice Field of Tibet. It diminished the output of the hydropower plants by flattening the Nepal-China Friendship Bridge which connects the two nations along with the highways. Furthermore, the flood also caused the death of numerous people along with disrupting trade and daily routines which showcase the severity of glitches which result from glacial hazards. The report validated the hypothesis which said the flood was a consequence of abnormal rainfall while in truth, it was a result of climate change accelerated climbing glacier melting. Within Nepal, India, and Peru the series of events demonstrates that mountainous regions around the world have become repeating the phenomenon of glacial lake bursting floods. The Rasuwagadhi GLOF exemplifies how climate change contributes to the expanding glacial lakes in the Himalayans and the increasing threats to the surrounding people and the structures.

There should be community drills to strengthen the proactive withstanding climate change assistance as a country. More so, a trilateral agreement should be reached to allow Hotak, China, and Nepal, to provide instantaneous data for monitoring and response over shared glacial hazards. Systematic monitoring and obstruction should be upheld so that relationships can be fostered while reducing risk. Moving forward, this study suggests that without preparedness measures for GLOF occurrences and cross-border partnerships, GLOF activity will continue to pose risks to life, infrastructure and the economy of the area.

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10. REFERENCES

- Carey, M. (2010). In the shadow of melting glaciers: *Climate Change and Andean Society*. Oxford: Oxford University Press.
- ECMWF. (2025, September 15). *European Centre for Medium-Range Weather Forecasts*. Retrieved from ECMWF Forecasts: <https://www.ecmwf.int/en/forecasts>
- GoogleEarth. (2025). *Satellite imagery of Rasuwagadhi, Nepal*. Retrieved from Satellite imagery of Rasuwagadhi, Nepal: <https://earth.google.com>
- Haeberli, W., Schaub, Y., & Huggel, C. (2017). Increasing risks related to landslides from degrading permafrost into new lakes in de-glaciating mountain ranges. *Geomorphology*, 293, 405-417. doi:<https://doi.org/10.1016/j.geomorph.2016.02.009>
- Huggel, C., Clague, J. J., & Korup, O. (2012). Is climate change responsible for changing landslide activity in high mountains? *Earth Surface Processes and Landforms*, 37(1), 77-91. doi:<https://doi.org/10.1002/esp.2223>
- ICIMOD. (2020). International Centre for Integrated Mountain Development. Retrieved from few glacial lake inventory report released, 47 potentially dangerous glacial lakes ranked: <https://www.icimod.org/new-glacial-lake-inventory-report-released-47-potentially-dangerous-glacial-lakes-ranked/>
- ICIMOD. (2025). *Preliminary analysis of the Rasuwagadhi flood: Evidence of supraglacial lake drainage*. International Centre for Integrated Mountain Development.

Ives, J. D., Shrestha, R. B., & Mool, P. K. (2010). *Formation of Glacial Lakes in the Hindu Kush-Himalayas and GLOF Risk Assessment*. Kathmandu: ICIMOD. doi:10.53055/ICIMOD.521

Khanal, N. R., Mool, P. K., Shrestha, A. B., & Bajracharya, S. R. (2015). A comprehensive approach and methods for glacial lake outburst flood risk assessment, with examples from Nepal and Bhutan. *International Journal of Water Resources Development*, 31(2), 219-237. doi:<https://doi.org/10.1080/07900627.2014.994118>

NDRRMA. (2025). National Disaster Risk Reduction and Management Authority. Retrieved from National Disaster Risk Reduction and Management Authority, Government of Nepal: <https://ndrrma.gov.np/en/news/191>

NEA. (2025). National Electricity Authority. Retrieved from National Electricity Authority, Nepal: <http://www.nea.org.np>

NOAA. (2025, July). National Oceanic and Atmospheric Administration. Retrieved from National Oceanic and Atmospheric Administration: <https://www.noaa.gov/weather>

RSS. (2023). Rastriya Samachar Samiti. Retrieved from Rastriya Samachar Samiti: <https://www.rssnepal.org>

RSS. (2025). Rastriya Samachar Samiti. Retrieved from Kerung border before flood [Photograph]: <https://www.rssnepal.org>

Shugar, D. H., Jacquemart, M., Shean, D., Bhushan, S., Upadhyay, K., & Westoby, M. J. (2021). A massive rock and ice avalanche caused the 2021 disaster at Chamoli, Indian Himalaya. *Science*, 373(6552), 300-306. doi:<https://doi.org/10.1126/science.abh4455>

TheKathmanduPost. (2025). The Kathmandu Post. Retrieved from Rasuwagadhi flood destroys friendship bridge, disrupts Nepal–China trade: <https://kathmandupost.com>